OSCAR NOLEN

CONTACT
Location: Charlotte, North Carolina
Phone: 7046355160
Email: oscarnolen@gmail.com
LinkedIn: www.linkedin.com/in/oscar-nolen-846269153
GitHub: https://github.com/oscarno22

EDUCATION

University of North Carolina at Charlotte, Charlotte, NC
08/2024-12/2025

- Master of Science in Computer Science
- Concentration in AI, Robots, Gaming

Duke University, Durham, NC
08/2020-05/2024

- Bachelor of Science in Computer Science
- Bachelor of Science in Mathematics

EXPERIENCE

Agentic AI Software Engineer
Ally Financial, Charlotte NC
01/2025-Present

- I'm coding powerful agents to drive automation at Ally, using modern frameworks such as LangGraph, Model Context Protocol, and OpenAI Agents SDK.
- I'm also developing a platform for these hosted agents, encompassing authentication and compartmentalization of agent resources (MCP tooling, database, etc).
- Security is critical at Ally, so I'm currently exploring ways to protect customer information from LLMs, add authentication at the tool-level, and handle complete agent failures.
- Key Skills:
    - LangGraph (Python), multi-agent workflow design
    - LLM integration for structured output, decision making, and summarization
    - OCR models for document analysis
    - AWS infrastructure, Terraform deployments
    - Gitlab pipelines, core CI/CD principles
    - Okta for user authentication, ID-JAG token authorization
    - Model Context Protocol (MCP) for tooling
    - Load testing, Unit testing, Prompt engineering, LLM output evaluations
    - Model and application approvals process

Software Dev Engineer Intern
Amazon Web Services, Seattle WA
05/2024-08/2024

- I contributed to AWS User Notifications (a notifications system) by designing and implementing an asynchronous central logging service. This new service will help identify missing notifications and related complications in the data-plane.
- For this service, I architected a data logging system with AWS Lambda and DynamoDB, and I developed APIs in Kotlin for seamless retrieval of notification traces.
- I built the infrastructure with AWS Cloud Development Kit and integrated with CloudWatch for monitoring and debugging. Additionally, I constructed a frontend UX in React.

Software Dev Engineer Intern
Amazon Web Services, Seattle, WA
05/2023-08/2023

- I contributed to Amazon Connect, an omnichannel cloud contact center platform, by developing a resource that helps replicate customer service workflows across multiple accounts.
- This new resource acts as a reusable skeleton for customer service workflows, enabling efficient customization and deployment. I developed this resource in the existing control plane service, and I built CRUD APIs in Java to support creation and management.
- Additionally, I added logic to validate the JSON representation of this new resource, ensuring data accuracy and visibility. This project was also implemented alongside JUnit and Integration testing.

Amazon Technology Intern
Amazon Web Services, Matthews, NC
05/2022-08/2022

- I worked on compatibility improvements in Babelfish for Aurora PostgreSQL, a new capability that enables Aurora to understand commands written for Microsoft SQL Server.
- As a part of this project, I identified missing catalog objects necessary for common usage patterns in Visual Studio, SQL Server Management Studio, and other SQL clients.
- I also collaborated with the Babelfish engineering team for replacing said objects, attending design reviews, and making updates to the code base.

SKILLS

Languages: Python, Java, Kotlin, JavaScript, TypeScript, C, C++
GenAI: LangGraph Platform, OpenAI Agents SDK, Model Context Protocol (MCP)
Machine Learning: PyTorch, Numpy, Pandas, Scikit-learn, NVIDIA GPU
Database: SQLite, PostgreSQL, DynamoDB, MongoDB, SQL, RDS
Cloud: AWS Lambda, ECS, EC2, S3, CloudWatch, CloudFormation, Terraform
Frontend: React, Next.js, Node.js
Miscellaneous: GitLab, Linux, FastAPI, CI/CD, Dependency Injection, Docker, REST APIs

PROJECTS

Memorization in VAE Models
UNC Charlotte
08/2025-12/2025

- Studied memorization in variational autoencoders (VAEs) over image data
- Focused on memorization that arises from varying representation in the training data distribution
- Wrote a custom VAE to work with eye images from a Kaggle dataset that was originally presented to classify Diabetic Retinopathy. Found that high memorization scores (above the 95th percentile) typically represent images that are outliers from the rest of the dataset

Personal Website

- Next.js application with LLM integration: https://portfolio-dun-sigma-31.vercel.app/
- LLM (for resume explanation) is hosted behind a serverless AWS Lambda

RELEVANT COURSEWORK

- Parallel Computing
- Data Structures
- Computer Architecture
- Database Systems
- Topological Data Analysis
- Differential Geometry
- AI/Machine Learning Techniques
- Parallel Computing
- Linear Algebra
- Design and Analysis of Algorithms
- Discrete Mathematics
- Real Analysis